Node Talker Manual

Receive and transmit audio, setting by setting

Node Talker (called Node Listen before 0.6.0) puts one node in your phone's browser: listen with about half a second of delay, talk with push-to-talk, and connect or tune favorites. On Android, the optional Node Talker Mobile app keeps both going with the screen off. This manual follows the audio both ways, from the radio to your ear and from your mic to the air, and explains every setting along the way: where it acts, what it does, when to change it, and how to measure the result with the built-in meters, tests and Audio check.

RECEIVE · TRANSMIT · SETTINGS · LEVELS · MOBILE · TROUBLESHOOTING

COVERED BY THIS EDITION

TOOL	FILE	VERSION
Node Talker	`nodetalker_v0_9_22_20261025.py`	0.9.22
Node Talker Mobile	`NodeTalker-0.3.1-debug.apk`	0.3.1 (Android 8.0 and later; Node Talker 0.6.11 or later for app audio; sections 7.9 and 11.1 list what 0.9.22 adds that the app does not yet use)
Dashboard	`nodecontrol_pi02w_v9_3_94` / `nodecontrol_v9_3_89`	9.3.94-pi02w for DVSwitch through the Dashboard; 9.3.89 or later (both builds) for ASL, EchoLink and M17

Node Talker is a common tool: the same file runs on the Pi Zero 2 W and the full build. The rest of the suite is covered in the user manual, `docs/ASL-DVS_User_Manual.pdf`.

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How to read it

If you only want to listen, chapters 2 and 3 are enough. To see how the pieces fit together, read chapter 4 first. If you transmit, read chapters 6 and 8 before your first contact: most audio complaints come from levels that were never set. Chapter 7 is the reference to come back to, and chapter 11 covers the Android app. Each setting is marked with where it is kept:

MARK	MEANING
NODE	Kept on the Pi in `/etc/asl_dvs/nodetalker.conf`. Changes it for everyone who uses Node Talker on this node. Needs you to be logged in.
BROWSER	Kept in this browser only. Each phone, tablet or computer has its own, and changing it on one leaves the others alone. No login needed.

> **NOTE** Levels are given in **dBFS**: decibels below digital full scale. 0 dBFS is the loudest a sample can be; −6 dBFS is half that; −20 dBFS is a tenth. Speech is measured as an average (what you hear as loudness) and as a peak (the loudest instant, which is what clips).

PART I

Getting going

What Node Talker does, how to set it up on a node, and a tour of the page.

1. What Node Talker is
2. Install and first setup
3. The page

What Node Talker is

A web page served by the Pi on port 8993 that lets a phone, tablet or computer listen to one node, transmit through it, and connect or tune a short list of favorites. Any browser will do; on Android, the optional Node Talker Mobile app (chapter 11) also keeps receive and transmit going with the screen off.

RECEIVE
The node's audio, about half a second behind, cleaned up on the Pi (CTCSS and hum cut, levels evened out, squelch tails muted) and played in the browser with a live spectrum.

TRANSMIT
Push-to-talk from the phone's microphone, processed in the page (filter, leveler, noise gate) and sent to a private node, 1921, that Node Talker keeps linked to your node.

CONNECT
Up to 15 favorites: AllStar nodes, EchoLink stations, DMR, STFU, YSF, P25, NXDN and D-STAR talkgroups and M17 reflectors, on its own or through the Dashboard.

MEASURE
Live meters, a mic test, a test through the node, and an Audio check that watches a few minutes of real use and says what to change.

1.1 What it needs

ITEM	WHY
An ASL3 node (Asterisk with app_rpt)	Node Talker takes the node's audio from Asterisk and transmits through it.
Python 3.11 or later	As on Debian 12 and 13. Node Talker uses only the standard library.
A modern browser	Chrome, Firefox, Safari or Edge from the last few years. The microphone needs the `https://` page.
Optional: Node Talker Mobile	The Android app (8.0 and later). Receive and transmit in the background, and push-to-talk on a plain `http://` node (chapter 11).
Optional: DVSwitch and USRP2M17	For the digital favorites. Node Talker tunes them itself with `dvswitch.sh` and `USRP2M17.ini`.
Optional: the Dashboard (Node Controller)	When it runs on the node, every favorite goes through it, so it shows what is tuned.

1.2 How it fits on the node

Node Talker adds two things to `/etc/asterisk/rpt.conf`, each with a backup first: one `outstreamcmd` line in your node's stanza, which sends the node's audio to Node Talker, and a private node **1921**, which carries your transmissions in. Everything else stays as it was: the radio, the other links and DVSwitch hear and send exactly what they did before. Chapter 4 follows both paths end to end and takes the private node apart line by line.

Install and first setup

Ten minutes from nothing to a first test transmission.

2.1 Install

1. The installer (7.2.0 and later) asks whether to add Node Talker; instmon (2.58.0 and later) lists it with the other tools and updates it. By hand: `sudo python3 nodetalker_v….py --install`. Over Node Listen, it moves the settings, favorites and certificate across and retires the old service.
2. Open `http://<pi-address>:8993`. Node Talker moves you to `https://`. The certificate is made on the Pi, so the browser warns once: accept it.
3. Log in with the Pi's root password.
4. On Android you can use Node Talker Mobile instead of the browser: install it, add the Pi's address and log in on the page it shows (chapter 11).

2.2 Receive: send the node's audio here

1. Setup tab, open the **RX** card (Node receive). The table lists each node in `rpt.conf` and its `outstreamcmd`, if any.
2. Pick your node and press **Send its audio here**. That node becomes Node Talker's main node.
3. Press **Restart Asterisk** (it asks first; every link drops and reconnects).
4. Listen tab: within a few seconds the status line shows a green **feed** light, and audio plays when the node passes any.

> **CAREFUL** A node takes one `outstreamcmd`. If yours already has one (a Broadcastify feed, for example), Node Talker refuses rather than replace it.

2.3 Transmit: add node 1921

1. Setup tab, open the **TX: node 1921** card (Transmit node) and press **Add node 1921**. Node Talker adds a `[1921]` stanza (your node's template, a USRP channel, no hang time, telemetry, courtesy tone or station ID) and a `[nodes]` line, after a backup.
2. Press **Restart Asterisk**. Within 20 seconds the card's summary reads *Set up and linked*: Node Talker links 1921 to your node and relinks it whenever it drops.

2.4 First test transmission

1. Setup tab, **TX: node 1921** card, Microphone: press **Test mic**, talk for 4 seconds, and read the report (section 9.2). Adjust **Mic gain** until it says *Good*.
2. Press **Test through the node**. It transmits for 4 seconds, over the air and to every link, and plays back what the node sent out.
3. Listen tab: hold the push-to-talk button and make a short call.

The page

Three tabs along the bottom: Connect, Listen and Setup. The title bar and the tabs bar stay put; only the middle scrolls, and the Listen tab is sized to fit the screen without scrolling.

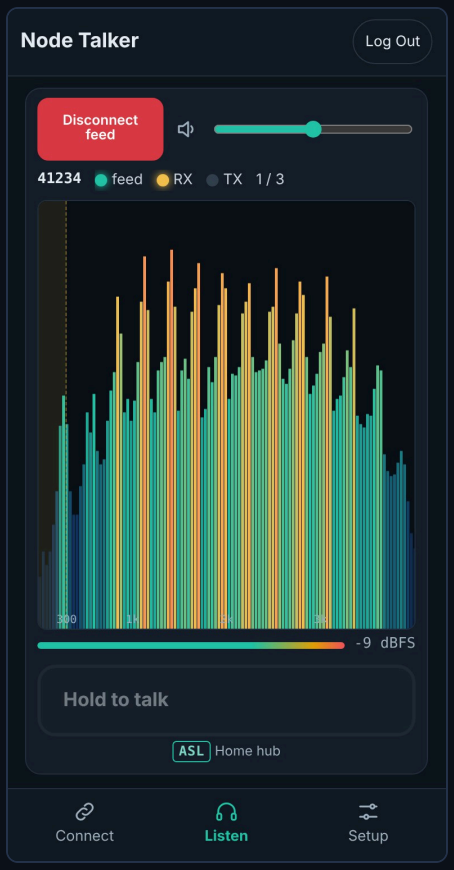


Figure 3.1 · Screenshot (test signal). The Listen tab, with the spectrum as bars and the link line under push-to-talk.

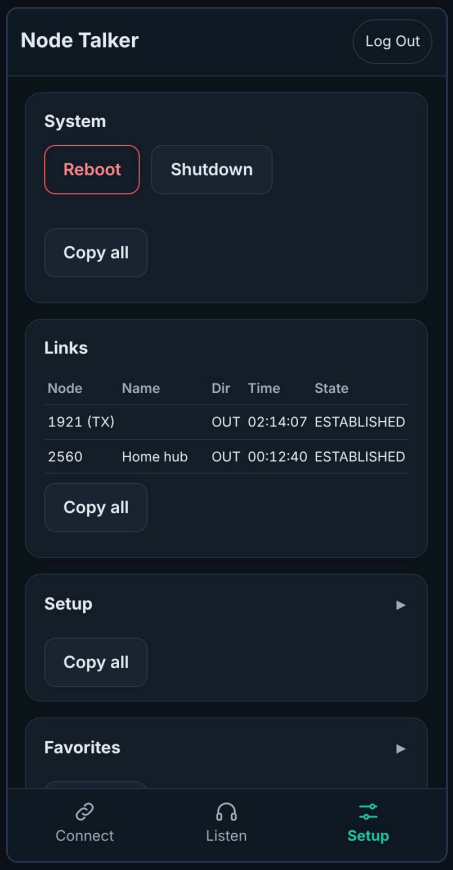


Figure 3.2 · Screenshot (test node). The Setup tab: System and Links at the top, then the cards, each closed with its summary and a Copy all button.

3.1 Title bar and login

PART	WHAT IT DOES
Version	Hidden on narrow phones.
Node Controller / InstMon	Opens the Dashboard, or switches to its tab if it is already open. When the Dashboard isn't running, the button opens InstMon instead.
Log In / Log Out	The root password. Five wrong tries from one address lock that address out for 5 seconds, doubling up to 5 minutes. A login lasts 12 hours from the last use.

With **Listen without logging in** on (Setup card), anyone who can reach the page gets the Listen tab without the password but can't transmit or change anything.

3.2 Listen tab, top to bottom

PART	WHAT IT DOES
Connect feed / Disconnect feed and volume	Starts and stops the audio (the button reads *Connect feed* while stopped and *Disconnect feed* while it plays; this is the page's own audio feed, not a link on the node); the slider beside it is the volume, 0 to 200 %. Listening starts by itself when the page opens (section 7.7).
Status line	Node number · **feed** light (green while audio arrives from Asterisk; "no feed 12 s" otherwise) · **RX** light (the node is passing audio) · **TX** light (someone transmits from Node Talker) · listeners now / limit · `CHECK 2:13` while an Audio check runs. Hover over a light for details.
Spectrum	0 to 4 kHz as bars (chapter 9). Off in Playback (RX card). With app audio in Node Talker Mobile it stays empty: the app plays the sound, not the page.
Level bar	The node's level, or your mic level (red **TX**) while you hold push-to-talk.
Message line	*Listening.*, *Tap anywhere to start listening*, or why the stream stopped.
Push-to-talk	Hold to talk, or tap to talk and tap to stop (chapter 6; Push-to-talk on this phone, TX card, picks which). Red **TALKING** with the time, or the guard countdown in tap mode; amber **Busy: who 0:12** when someone else transmits; a 3-2-1 count in the spectrum's corner while **Quick Key Guard** holds it.
Link line	Under the push-to-talk button: what the node is linked to right now, with a mode tag and a name (for example `ASL` and the net's name, or the talkgroup or reflector of a tuned DVSwitch or M17 bridge). It reads the node's own links, not your Connect or Tune choice, so a link made from the Dashboard or another phone shows too. Other states: *No ASL node linked*, *Bridge linked, nothing tuned*, *Not connected* and *Node unreachable*. Node 1921 and the bridge nodes are never shown as the link itself; a tuned DVSwitch or M17 bridge follows an ASL name as a tag. Names come from your favorites, then the AllStar list (`astdb.txt`).

3.3 Setup tab

At the top, always open: **System** (**Reboot** and **Shutdown** the Pi, each asked first) and **Links** (the node's links, one per line; node 1921 is marked TX). Below them, eight cards that follow the audio from one-time setup to everyday use: **Setup** (the node's settings, Save, Restart Asterisk), **Favorites**, **Digital**, **RX** (Node receive, Playback, Audio levels and RX cleanup), **TX: node 1921** (Transmit node, Audio levels, Push-to-talk on this phone, Microphone, Headset / audio output), **Audio check** (with the Tone test), **Phone** and **Log**. Tap a title to open or close it; every card starts closed, showing a one-line summary such as *Set up and linked* or *300 Hz cut · auto level · gate light*. Each card ends with a **Copy all** button that copies the whole card as plain text, folded sections and field values included (password fields are left out), which is the easiest way to show someone your settings. Chapter 7 goes through them all.

3.4 Connect tab

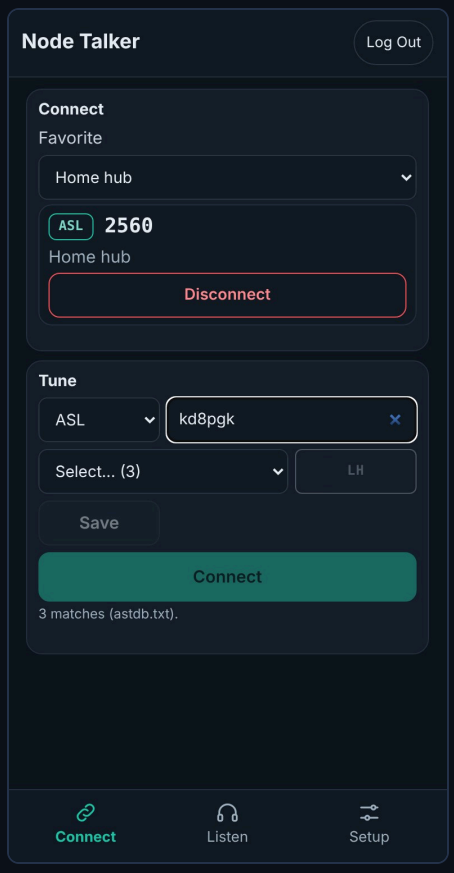


Figure 3.3 · Screenshot (test node). The Connect card with a linked favorite, and the Tune card searching AllStar nodes.

Connect: pick one of your favorites and press **Connect**; the button turns into **Disconnect** while it is on.

Tune: anything that isn't a favorite. Pick the mode (ASL, EchoLink, DMR, STFU, YSF, P25, NXDN, D-STAR, M17), search or type, then **LH** (Last Heard), **Save** as a favorite or **Connect**, which turns into **Disconnect** while what you picked is linked. Chapter 10 has both.

The node's links are on the Setup tab (Links card).

PART II

The audio, end to end

Where the sound goes, what changes it at each step, which setting acts where, and how to set and check levels.

4. The RX and TX chain, and the private node
5. Receive (RX) audio
6. Transmit (TX) audio
7. Settings reference
8. Getting levels right
9. Reading the spectrum and the meters

The RX and TX chain, and the private node

The whole loop on one page: how the node's audio reaches your phone, how your voice gets back on the air through a private node, and what each piece is. Chapters 5 and 6 then go through each direction step by step.

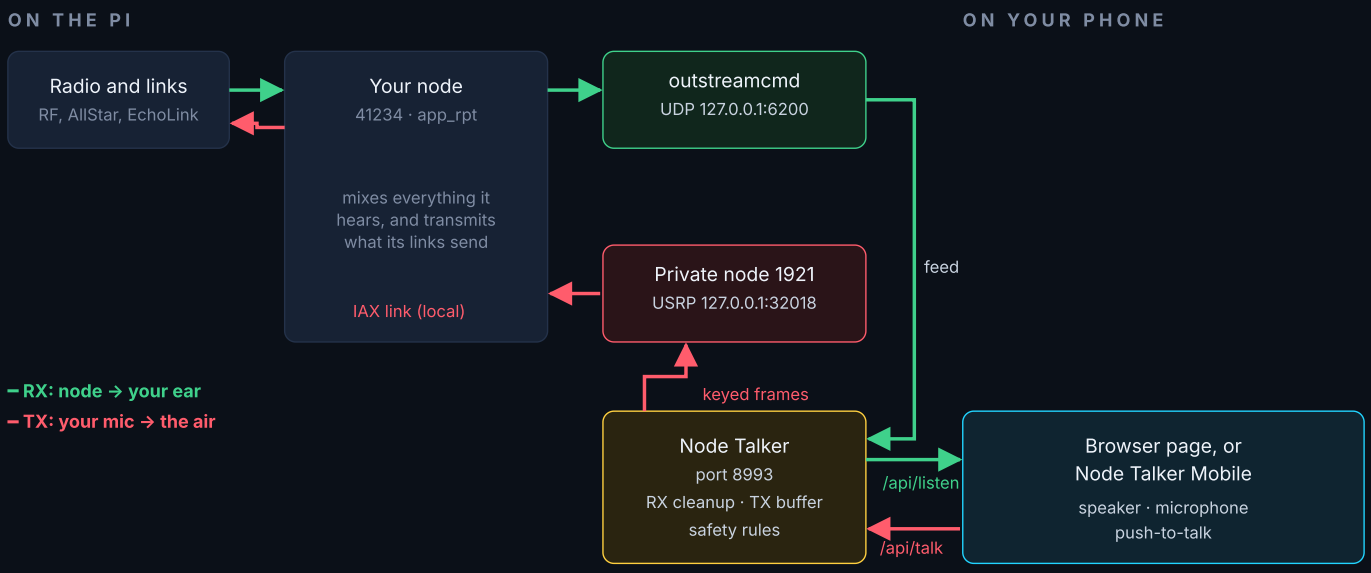


Figure 4.1 · The loop. Receive (green) leaves the node through `outstreamcmd`; transmit (red) comes back in through private node 1921, which is linked to your node like any other link, so your voice goes out on the radio and to every link.

4.1 The pieces

PIECE	WHAT IT IS	WHERE IT IS SET UP
Your node	The node in `rpt.conf` you picked with **Send its audio here** (41234 in the examples). Node Talker calls it the main node.	RX card (Node receive)
`outstreamcmd`	One line in your node's stanza. app_rpt starts the program it names and writes everything the node hears into it: 8,000 samples a second, 16 bits. The program is a small part of Node Talker that passes the audio on over UDP.	Receive card (writes the line)
Node Talker	The service on port 8993: the page, the audio pump, RX cleanup, the TX buffer and every safety rule. It sends the audio to each listener and takes transmit audio from one talker at a time.	Settings, Audio levels
The page, or the app	Plays the audio and records your microphone. In a browser the page does both; in Node Talker Mobile with app audio, the app does them and the page only shows what happens.	Microphone, Playback; the app's Audio settings
Private node 1921	A node of its own in `rpt.conf` with no radio: its "radio" is a USRP channel that Node Talker feeds. When Node Talker sends keyed audio to it, 1921 transmits, and because it is linked to your node, your node passes the audio to its radio and links.	Transmit card (writes the stanza)
The link 1921 → your node	An ordinary local IAX link (`ilink 3`, transceive). Node Talker makes it after Asterisk starts and checks it every 20 seconds, relinking if it drops.	Automatic

4.2 What happens when you press push-to-talk

1. **Press.** The page (or the app) sends `{"op":"key"}` on the talk channel, the WebSocket `/api/talk`, and starts the microphone at once. Audio captured before the answer is held (up to half a second) so the first word isn't lost.
2. **Checks on the Pi.** Node 1921 must be set up and linked; nobody else may be transmitting from Node Talker; with the busy lockout on, the node must be quiet (or your own unkey less than 1 s ago). Otherwise the answer is `denied` with the reason, and the page says why.
3. **Keyed.** The answer `{"ev":"keyed","tot":120}` carries the TX time-out. The page turns red and counts; the held audio and every 20 ms frame after it go to the Pi.
4. **On the Pi.** Frames collect in the TX buffer (120 ms by default), get TX gain if it is on, and leave every 20 ms as USRP packets with the key flag set, to `127.0.0.1:32018`.
5. **On the air.** Node 1921 hears a keyed channel, so it "receives"; the link carries that to your node, which keys its radio and sends the audio to every other link. Listen audio on your phone is muted meanwhile, and comes back 1 s after you let go.
6. **Release.** `{"op":"unkey"}`. The Pi plays out what is left in the buffer, fades the last frame, sends three unkey packets, and answers `{"ev":"unkeyed","why":"release"}`.
7. **Or it stops by itself:** at the TX time-out (`timeout`), when no audio has arrived for 1 s (`deadman`), when the talk channel closes (`closed`), or at the guard timer in tap mode.

4.3 And on receive

Everything your node hears, including your own transmission as the node passes it, goes out through `outstreamcmd` to Node Talker, which cuts it into 20 ms frames, cleans it up once for everyone and streams it to each listener over `/api/listen`. Your phone buffers a quarter of a second and plays it. Chapter 5 has the details and the timings.

4.4 Example: the private node 1921, line by line

A **private node** is a node number from 1 to 1999. These numbers aren't registered with AllStarLink, so nobody outside can connect to them: they exist only on your Pi, linked locally. That makes them the right tool for feeding audio into a node from software. Node Talker uses 1921 and writes this when you press **Add node 1921** on the TX card, Transmit node (here for a node 41234 that uses ASL3's usual `node-main` template):

```
; nodelisten: transmit node for the Node Listen page (remove it from that page)
[1921](node-main)
rxchannel = USRP/127.0.0.1:34018:32018
duplex = 0
hangtime = 0
althangtime = 0
holdofftelem = 1
telemdefault = 0
nounkeyct = 1
idrecording = |t(0,0,10,0)

[nodes]
41234 = radio@127.0.0.1/41234,NONE
1921 = radio@127.0.0.1/1921,NONE
```

LINE	WHY IT IS THERE
`; nodelisten: …`	The marker Node Talker (and the older tools) look for. Without it, Node Talker treats a `[1921]` as someone else's and won't touch it.
`[1921](node-main)`	The node, built on the same template as your node, so it gets the same defaults (codecs, timers, functions).
`rxchannel = USRP/127.0.0.1:34018:32018`	Its "radio" is a USRP channel on this Pi: it takes audio on UDP 32018 (from Node Talker) and would send on 34018. Nothing leaves the Pi.
`duplex = 0`	Half duplex, like a simplex radio: it doesn't repeat what it hears back out.
`hangtime = 0`, `althangtime = 0`	No hang after you unkey, so the node drops straight away.
`holdofftelem = 1`, `telemdefault = 0`	No telemetry (IDs, link announcements) from 1921 itself; your node keeps its own.
`nounkeyct = 1`	No courtesy tone when you unkey from Node Talker.
`idrecording = |t(0,0,10,0)`	Node 1921's own station ID is a 10 ms silent tone. Without this line 1921 would use the ID of the template it is built on, which on a stock ASL3 install is someone else's callsign, and send it on the air whenever it keys.
`1921 = radio@127.0.0.1/1921,NONE`	The `[nodes]` entry that lets your node find 1921 on this Pi. If Asterisk listens on another IAX port than 4569, the port is added (`127.0.0.1:4570`).

Checking it from the Pi

After **Restart Asterisk**, these show that 1921 exists and is linked:

```
sudo asterisk -rx "rpt lstats 41234"
NODE      PEER                RECONNECTS  DIRECTION  CONNECT TIME        CONNECT STATE
----      ----                ----------  ---------  ------------        -------------
1921      127.0.0.1           0           OUT        02:14:07:310        ESTABLISHED

sudo asterisk -rx "rpt nodes 1921"

************************ CONNECTED NODES ************************

T41234
```

To link it by hand (Node Talker does this by itself within 20 seconds): `sudo asterisk -rx "rpt cmd 41234 ilink 3 1921"`. To see it key while you talk: `sudo asterisk -rx "rpt stats 1921"` shows *Signal on input* as *YES*. The *T* in front of 41234 means a transceive link: audio goes both ways.

> **NOTE** Node Talker 0.9.22 writes the `idrecording` line only into a new stanza. A 1921 that an earlier version added keeps its stanza as it is: add the line by hand, or press **Remove** and then **Add node 1921** on the TX card (and Restart Asterisk).

> **CAREFUL** Node 1921 is fixed. If your `rpt.conf` already has a `[1921]` for something else, the TX card refuses rather than overwrite it: move the other one to a free private number (for example `[1922]`, with its `[nodes]` line) first.

> **NOTE** Private nodes follow their own rules elsewhere too: connecting a favorite drops every other *public* link but never node 1921, the digital bridge nodes or any other private node (chapter 10).

Receive (RX) audio

From the radio to your ear: everything the node hears, cleaned up once on the Pi for every listener, then buffered, played and drawn by each phone.

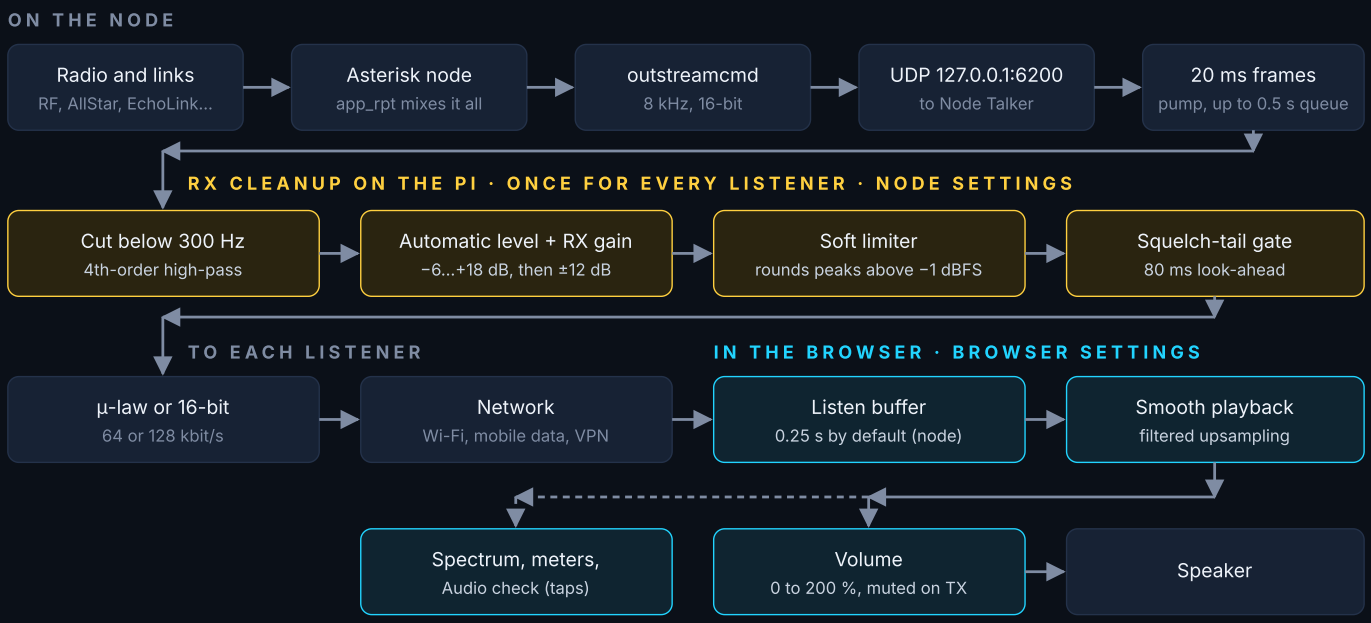


Figure 5.1 · The receive path. Amber steps run on the Pi once for everyone, blue steps on each listener's own device. The dashed branch is measuring only: the spectrum, the level bar and the Audio check see the audio before your volume.

5.1 From the node to Node Talker

app_rpt mixes everything your node hears (the radio, every link, telemetry) and writes it, 8,000 samples a second at 16 bits, to the `outstreamcmd` program, a small part of Node Talker that passes it on to `127.0.0.1:6200`. The audio pump cuts it into 20 ms frames (160 samples) and sends one to every listener every 20 ms. If audio arrives in bursts, up to 0.5 s waits in a queue; beyond that the oldest is dropped and counted.

The status line's **feed** light is green while frames arrive. The **RX** light and the busy lockout use the node's own level before any cleanup: a frame louder than −50 dBFS counts as audio, held for 0.6 s.

5.2 RX cleanup on the Pi

Three steps, done once for every listener, change only what Node Talker's listeners hear. They are on by default; chapter 7 has each setting in detail. Nothing is done while nobody listens, and near-silent frames (under −65 dBFS) skip the work.

STEP	WHAT IT DOES TO THE SOUND
Cut below 300 Hz	Removes hum and the lower CTCSS (PL) tones: 100 Hz drops by about 38 dB, 150 Hz by 24 dB, 250 Hz by 7 dB. Voice above 300 Hz is untouched.
Automatic level	Aims speech at about −20 dBFS: brings quiet stations up by as much as 18 dB and loud ones down by as much as 6 dB. It turns down within a frame or two, comes up at 3 dB a second (faster for the first half second of a new transmission), and holds still between transmissions. RX gain (Audio levels) is added on top.

Soft limiter	Up to about −1 dBFS nothing changes; above that, peaks are rounded off smoothly instead of clipped.
Squelch-tail gate	Mutes the burst of hiss after a station unkeys, and long silences. It tells hiss from voice by how alike neighbouring samples are: voice scores high, hiss low. A frame counts as hiss only when its samples one and two apart are both unrelated (flat hiss), so steady tones and loud "s" sounds are not muted. To mute a tail from its first frame, it holds the audio back 80 ms; when the gate opens it keeps the 40 ms before the opening and fades in, so the start of a word is not clipped.

5.3 To your phone and out of the speaker

- **Format:** 16-bit (128 kbit/s) by default, or µ-law (64 kbit/s) with **16-bit audio** off. µ-law is the phone-line format: half the data, a little more hiss on quiet audio.
- **Listen buffer:** the page plays each 100 ms of audio a fixed time after it arrives, 0.25 s by default, so a late packet doesn't make a gap. When audio arrives later than the buffer can cover, playback restarts (a *dropout*), and the Audio check counts it.
- **Smooth playback:** the page upsamples 8 kHz to the phone's own rate (44.1 or 48 kHz) with a proper filter, which removes the faint sizzle some phones add.
- **Volume** and then the speaker. The spectrum and meters see the audio before the volume.

5.4 Where the half second goes

STAGE	DELAY
Asterisk to Node Talker	a few tens of ms
Squelch-tail gate look-ahead	80 ms (only with the gate on)
Page chunks	up to 100 ms
Listen buffer	250 ms by default (150 ms to 1 s)
Network	a few ms on Wi-Fi at home; more on mobile data

The total is about half a second at home. This is why the node's own audio comes back late when you test through it, and why Node Talker ignores the busy lockout for 1 s after your own transmission.

Transmit (TX) audio

From your mic to the air: shaped in the page, sent in 20 ms pieces, smoothed on the Pi and fed to node 1921, which is linked to your node.

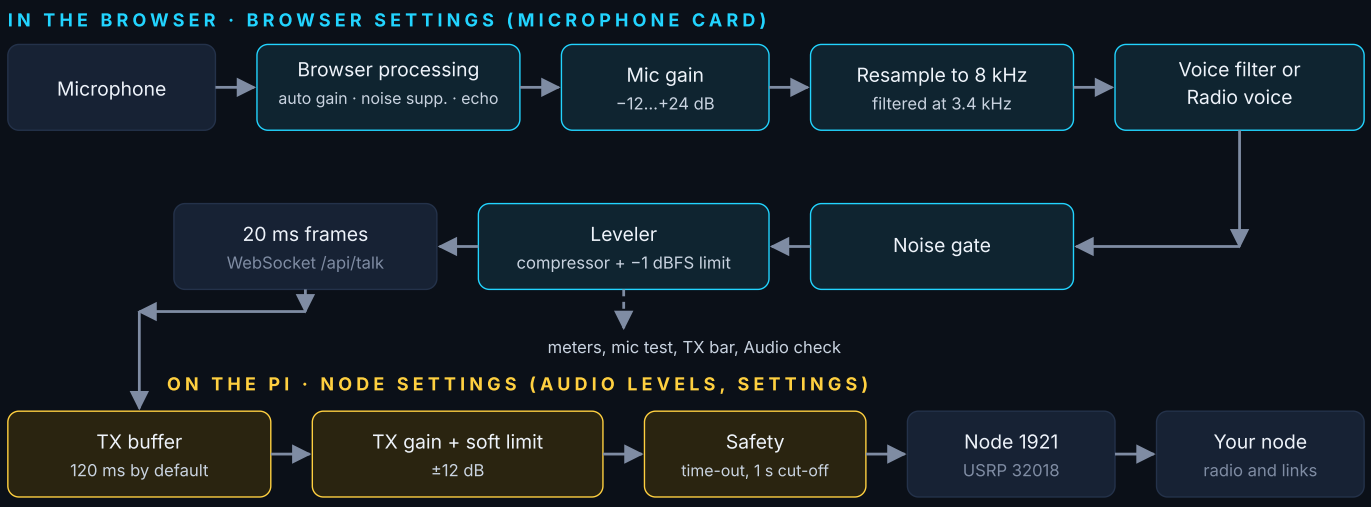


Figure 6.1 · The transmit path. Blue steps run in your browser with your device's settings, amber steps on the Pi with the node's settings. The meters measure right after the leveler, so they show what the Pi will get.

6.1 In the browser

1. **Browser processing.** The phone's own **Automatic gain**, **Noise suppression** and **Echo cancellation**, each switchable. They run before Node Talker sees the sound.
2. **Mic gain**, −12 to +24 dB: a plain volume control on the mic.
3. **Resample to 8 kHz** with a filter that cuts at 3.4 kHz, so sounds above it don't fold back as hiss or whine.
4. **Voice filter** (cuts rumble below about 250 Hz) or **Radio voice** (cuts below 300 Hz and lifts 2.5 kHz by 4 dB, a crisper, more "radio" sound). With Radio voice on, the voice filter is skipped.
5. **Noise gate** (Off, Light, Strong): turns the mic down between words.
6. **Leveler**: a gentle compressor that keeps your level even, followed by a limiter that never lets a peak past −1 dBFS.
7. Every 20 ms (160 samples) a frame goes to the Pi over the WebSocket `/api/talk`.

6.2 On the Pi and on the air

1. **TX buffer.** The Pi waits until it has this much of your audio (120 ms by default) before it starts sending, so short hiccups on the phone's connection don't make gaps. If it runs dry anyway, it sends silence, refills, and counts a *gap*.
2. **TX gain** (−12 to +12 dB, with a soft limit at about −1 dBFS), only with **Adjust audio levels here** on.
3. The frames go as USRP packets to node 1921 (`127.0.0.1:32018`), keyed. On release the last frame is faded out and three unkey packets follow. If the audio has a gap, the last frame is repeated faded out instead of a hard cut, and the frame after the gap fades in.

4. Node 1921 is linked to your node, so your audio goes out on the radio and to every link, like a local user's.

6.3 The safety rules

RULE	WHAT HAPPENS
Busy lockout	With **Don't transmit while the node is busy** on, a press is refused while the node passes audio (the RX light), except within 1 s of your own unkey.
One at a time	A second person's press is refused while someone transmits from Node Talker.
Time-out	The transmission stops at **TX time-out** (120 s by default). Let go and press again.
Audio stopped	If no audio reaches the Pi for 1 s (a dropped connection), it unkeys.
Guard timer	In tap-to-talk mode (and for the headset button), the transmission stops after the guard time (30–180 s, 60 s by default; never past the TX time-out), with two short beeps and a flashing count 10 s before.
Quick Key Guard	With **Quick Key Guard** on (it was called Wait 4 s after receive), push-to-talk is locked while the node receives and for 4 s after, so other stations can answer first.
Page hidden	In a browser, switching apps, locking the phone or leaving the page releases push-to-talk. In Node Talker Mobile a tap-to-talk transmission carries on and hold-to-talk releases (chapter 11).

Settings reference

Every setting, card by card, in the order of the audio (the Favorites and Digital cards are in chapter 10). NODE settings need you to be logged in and apply to everyone; BROWSER settings are this device's own.

7.1 Node receive NODE RX card

CONTROL	WHAT IT DOES
Node table	Each node in `rpt.conf` and its `outstreamcmd`: *Sends audio here*, *to another UDP port (press Send again)*, another program's line, one inherited from a template, or *Not sending*.
Send its audio here	Adds Node Talker's line to the picked node (taking it off any other node), backs up `rpt.conf`, and makes that node the main node. Takes effect after Restart Asterisk.
Remove	Takes the line out again (with a backup).
Restart Asterisk · Refresh	Restart asks first; every link drops and reconnects. Refresh rereads `rpt.conf`.
Line added · Feeder · UDP input	The exact line; whether Asterisk is running the feeder program now; whether Node Talker is listening on the UDP port.

7.2 Transmit node and push-to-talk TX card

CONTROL	WHAT IT DOES
State	*Not set up*, *Set up and linked*, *Set up; linking*, *Asterisk does not know it yet: restart Asterisk*, or what is wrong.
Add node 1921	Adds the `[1921]` stanza and its `[nodes]` line (with your IAX port if it isn't 4569); also repairs a wrong `[nodes]` line. Refuses if `rpt.conf` has a `[1921]` it didn't add.
Remove	Takes both out (asks first). Push-to-talk stops working.

Push-to-talk on this phone BROWSER

SETTING	DEFAULT	WHAT IT DOES, AND WHEN TO CHANGE IT
Hold to talk / Tap to talk, tap to stop	Hold	Tap mode: one tap keys, the next unkeys; a tap within 0.5 s of keying is ignored. Space or Enter work the same way.
Guard timer	60 s	30, 60, 90, 120 or 180 s, for tap mode and the headset button (section 6.3).
Quick Key Guard	off	Locks push-to-talk while the node receives and for 4 s after, counting 3 to 0 in the corner of the spectrum, so other stations can answer first. It was called Wait 4 s after receive.

Save on this phone also keeps a copy on the Pi (`ptt_mode`, `ptt_guard_sec`, `ptt_rx_gap`); a phone with nothing saved starts from that copy, and **Restore from the Pi** puts it back. The guard timer never runs past the node's TX time-out.

7.3 Setup card NODE

SETTING	DEFAULT	RANGE	WHAT IT DOES, AND WHEN TO CHANGE IT
Web port	8993	1024–65535	The page's port. Change only if something else uses 8993. Restarts Node Talker.
UDP port	6200	1024–65535	Where the node's audio arrives. Changing it rewrites the `outstreamcmd` line too; restart Asterisk.
Max listeners	3	1–10	How many people can listen at once. Each takes 64 or 128 kbit/s of upload on your internet line when they're away from home.
TX time-out (s)	120	30–600	The longest one transmission can last.
HTTPS	on	on / off	Needed for the microphone. Off only for a listen-only node. Restarts Node Talker.
Don't transmit while the node is busy	on	on / off	The busy lockout (section 6.3). Turn it off only if a constant noise on the node keeps it busy.
Listen without logging in	off	on / off	Anyone who can reach the page may listen (not transmit).

Tune card on this phone: sliders for the Tune card's gap, height and text size (kept in this browser), with **Default spacing**. **Save** keeps the node settings above; **Restart Asterisk** asks first and drops every link while Asterisk comes back.

7.4 Audio check card BROWSER

Length (2, 5, 10 or 15 minutes; 5 by default), **Start check**, **Stop**, **Copy all** and **Prompt me to talk during the check** (on by default). The report has the Receive and Transmit tables, a **Per minute** strip, **What to change** with **Apply**/**Undo**, and the buttons **Compare with last check**, **Copy report**, **Download report** and **Compare with a saved report**. The **Tone test** is further down the same card. Section 8.4 explains the check and section 8.5 the tone test.

7.5 Audio levels NODE RX and TX cards

The settings are split by direction: **Adjust audio levels here**, **Mic to node (TX)**, **TX buffer** and the reset button are on the TX card; **Node to phone (RX)**, **Listen buffer** and RX cleanup are on the RX card. Each card has its own **Save**.

Adjust audio levels here

Off by default. Off, the four settings below are not used: no TX or RX gain, a 0.25 s listen buffer and a 120 ms TX buffer. On, they take effect. The RX cleanup further down works either way.

SETTING	DEFAULT	RANGE	WHAT IT DOES, AND WHEN TO CHANGE IT
Mic to node (TX)	0 dB	−12 to +12 dB	Gain on every transmission from Node Talker, on the Pi, for everyone. Use it when all phones come out too quiet or too loud on the air (the node test shows it). For one phone, use its Mic gain instead.
Node to phone (RX)	0 dB	−12 to +12 dB	Gain on what every listener hears, after Automatic level. Use it when audio is quiet even with Automatic level on, or when you turn Automatic level off.

Listen buffer	0.25 s	0.15, 0.25, 0.5, 1 s	How far behind real time the page plays. Larger rides over a weak connection without dropouts; smaller is closer to live. 0.15 s on good Wi-Fi, 0.5–1 s on mobile data.
TX buffer	120 ms	60, 120, 200, 300 ms	How much of your audio the Pi collects before sending. Raise it if transmissions have gaps (TX gaps, the Audio check); lower it for a slightly quicker start.

Below them: **TX peak** (the loudest point of the current or last transmission, after TX gain), **RX level** (the node's level, before cleanup) and **TX gaps**. Aim for TX peaks of −6 to −3 dBFS on normal speech.

RX cleanup

SETTING	DEFAULT	WHAT IT DOES, AND WHEN TO CHANGE IT
Cut below 300 Hz	on	Removes hum and CTCSS tones from what listeners hear (section 5.2). Off only to hear the tones themselves, for example to check which one a station sends. It costs the most of the three on the Pi.
Automatic level	on	Evens out quiet and loud stations. Off to hear stations at their real levels, or for an exact node test.
Squelch-tail gate	Light	**Light** mutes the hiss after a station unkeys and long silences. **Strong** also mutes shorter and quieter hiss: use it on a node with noisy links, and turn it back if the start of speech is ever clipped. **Off** passes everything, with no 80 ms delay.

Processing load: how much of one CPU core the cleanup took over the last 5 seconds. On a Pi Zero 2 W it should stay at about 10 % or less while audio passes. **Save** keeps everything on the card; **Reset gains and buffers to defaults** (TX card) puts them back (Save to keep it). A favorite can carry its own RX and TX gain (section 10.1); while it is on, those values replace the two gains above, even with **Adjust audio levels here** off.

7.6 Microphone BROWSER TX card

SETTING	DEFAULT	WHAT IT DOES, AND WHEN TO CHANGE IT
Leveler	on	A compressor and limiter in the page: quiet speech comes up (by up to 9 dB), loud speech is held back, and nothing passes −1 dBFS. Leave on; turn off only if you level the audio somewhere else.
Automatic gain	off	The browser's own leveling. It fights the leveler and undoes Mic gain, so leave it off when the leveler is on.
Noise suppression	on	The browser removes steady background noise (fans, traffic). Turn off if your voice sounds watery.
Echo cancellation	off	Rarely needed: listen audio is muted while you talk. Turn on only with a speaker right next to the mic.
Voice filter	on	Cuts rumble below about 250 Hz (handling noise, wind, desk thumps).
Radio voice	off	Cuts below 300 Hz and lifts 2.5 kHz for a crisper sound that cuts through on FM. Leave off if your audio already sounds bright, or if the radio adds pre-emphasis.
Noise gate	Off	**Light** opens above −45 dBFS and turns the mic down 15 dB between words; **Strong** opens above −35 dBFS and turns it down 30 dB. Use Light in a noisy room. If the starts of words are clipped, go back a step (the Audio check measures it).

Mic gain	0 dB	−12 to +24 dB. The main control for your level: set it with the mic test (section 8.2).

Test mic and **Test through the node** are in this section of the TX card; chapter 9 shows their meters and reports.

7.7 Playback BROWSER RX card

SETTING	DEFAULT	WHAT IT DOES, AND WHEN TO CHANGE IT
Show the spectrum	on	The bars on the Listen tab. Off saves battery on an old phone.
Start listening when the page opens	on	Plays as soon as the page loads or you log in. Most browsers need one tap first; the page asks for it.
Keep the screen on	off	Shown where the browser supports it: keeps the screen awake while you listen or talk.
16-bit audio	on	Cleaner on quiet audio, 128 kbit/s instead of 64. Turn off on a slow or metered connection.
Smooth playback	on	Filtered upsampling, no sizzle. Turn off only on a very slow device.

7.8 Log card

What Node Talker has done: listeners, logins, transmissions (with length, gaps and why they ended), connects and every `rpt.conf` edit. Fetched only while the card is open.

7.9 Phone card BROWSER

Read-only: nothing here changes a setting. It lists what this phone's browser can tell the page about audio, so you can see why a setting has no effect. **Refresh** rereads it, and **Copy all** puts it in a message when you ask someone for help.

- Whether the page is secure (`https://`; the microphone and device names are blocked if not), the network state, the sound system's rate and playback latency, and whether the browser can pick the output.
- The output in use, the microphones and output devices found, and the microphone in use (names appear once the microphone has been allowed).
- Which of echo cancellation, noise suppression and automatic gain the browser can do, and the page's own Leveler, Automatic gain, Noise suppression, Echo cancellation, Voice filter / Radio voice, Noise gate and Mic gain.
- A star (*) marks a value you can change on the phone. A page can't read the phone's media volume, so that row says so unless the app reports it.
- Inside Node Talker Mobile a second table, *Node Talker Mobile*, shows the app version and whether the audio is app audio or page audio, and rows the app doesn't use are marked *not used: app audio*. The app's own settings (output, receive buffer, echo and noise filter, microphone rate, phone volume, battery saver, keep the screen on) appear once the app reports them; Node Talker Mobile 0.3.1 does not yet, and the table says so.

7.10 Settings that are grayed out

A setting that can't do anything right now is grayed out, and a note under its section says why, so you don't spend time on a control that has no effect. The page rechecks every 2 seconds.

GRAYED OUT	BECAUSE
The page's playback options, its microphone processing, Test mic, Test through the node, output and microphone picking, the headset button and Start check	The app is doing the audio (Node Talker Mobile with app audio).
The microphone options	The page isn't the `https://` one, so the browser blocks the microphone.
Echo cancellation, Noise suppression, Automatic gain	This browser can't do them.
Voice filter	Radio voice is on, which includes it.
The gain and buffer fields	Adjust audio levels here is off.

Getting levels right

A short procedure for each direction, in the order that avoids fighting yourself. Do it once when you set the node up and again when you change phones, mics or radios.

8.1 Who hears which gain

GAIN	WHERE	WHO IT CHANGES
Radio RX / TX levels, DVSwitch gains	SysMon, Tune tab; DVSwitch	Everyone: the air, every link, Node Talker.
Automatic level, RX gain, Cut below 300 Hz, gate	Node Talker, Audio levels NODE	Node Talker's listeners only.
TX gain	Node Talker, Audio levels NODE	Transmissions from every Node Talker user.
Mic gain, leveler, filters, gate	Node Talker, Microphone BROWSER	Transmissions from this device only.
Volume	Listen tab BROWSER	What you hear on this device only.

TIP Fix things where they start. If the node's own audio is wrong on the air and on links, fix it in SysMon (radio) first; Node Talker's settings only paper over it for Node Talker's users.

8.2 Transmit, step by step

1. **Start clean.** Microphone (TX card): Leveler on, Automatic gain off, Noise suppression on, Voice filter on, Noise gate Off, Mic gain 0 dB.
2. **Test mic.** Hold the phone where you will hold it on the air and talk normally for 4 seconds. Targets: speech **−24 to −12 dBFS**, peaks **−8 to −1 dBFS**, limiter working under 20 % of the time. Change Mic gain by the amount the report suggests and test again until it says *Good*.
3. **Noise.** If the report says *Noisy*, move to a quieter spot or closer to the mic first; then try Noise gate Light.
4. **Test through the node.** For an exact comparison, turn Automatic level off for a minute (Audio levels). *Came back* should be within about 6 dB of *Sent*. If every phone comes back quieter or louder by the same amount, use TX gain; if the radio side is off, use SysMon's Tune tab. Turn Automatic level back on.
5. **On the air.** Run an Audio check during a real contact or net (section 8.4) and apply what it suggests.

8.3 Receive, step by step

1. **The node first.** With the radio's own levels right (SysMon, Tune), the RX level in Audio levels (RX card) should peak around −10 to −3 dBFS on a strong local station.
2. **Look at the spectrum.** A steady line or band below 300 Hz is a CTCSS tone or hum; with Cut below 300 Hz on it should be faint or gone. Thin bright lines right after transmissions are squelch tails: the gate should remove them.

3. **Leave Automatic level on.** If speech is still quiet (the Audio check says so), raise RX gain by the amount it suggests.
4. **Dropouts.** If audio breaks up on your connection, raise the Listen buffer a step.

8.4 The Audio check

The quick tests show four seconds; the Audio check shows a few minutes of real use, with several stations and several of your own transmissions.

1. Setup tab, **Audio check** card: choose a length and press **Start check**. The screen is kept awake while a check runs (a warning shows if the browser can't).
2. Go to the Listen tab and use the node as usual: listen, talk, listen. The status line shows `CHECK 2:13`; tap it to come back. With **Prompt me to talk during the check** on (the default), the page asks you to talk at 0:30 and every 2 minutes after, so the transmit side has real speech to measure.
3. When it ends (or you press Stop), open the card. Each row is green, amber or red against its target, and **What to change** lists suggestions with amounts, ordered by how much audio they affect: lost or distorted audio first, then levels, then noise and filters. At most two carry **Apply** at once; the rest are marked *Later*. Level suggestions wait for 5 seconds of speech.
4. Press **Apply** on a suggestion to make the change (**Undo** puts it back), run another check, and press **Compare with last check** to see the difference.

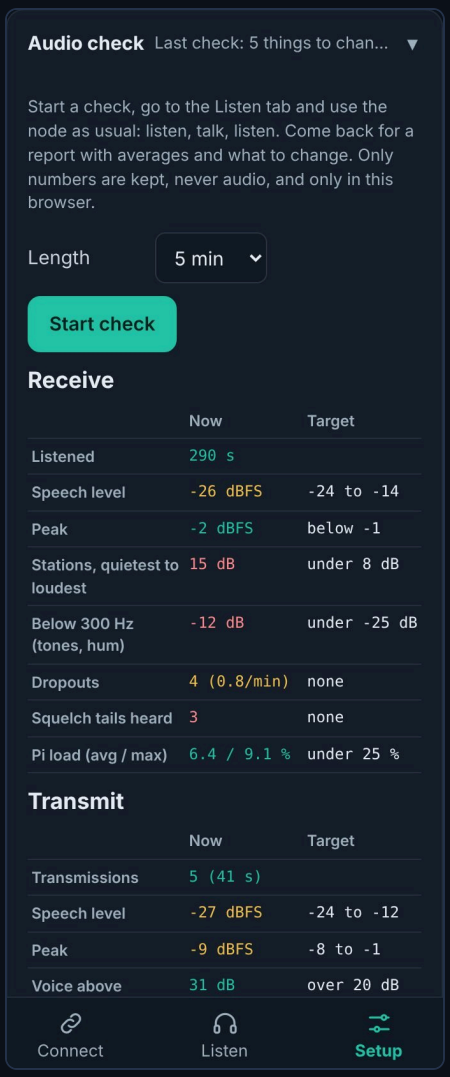


Figure 8.1 · Screenshot (example data). An Audio check report: stations 15 dB apart, hum below 300 Hz, dropouts and squelch tails, each with a suggestion below.

ROW	TARGET	TYPICAL FIX
RX speech level	−24 to −14 dBFS	Automatic level; RX gain
RX peak	below −1 dBFS	Lower RX gain, or Automatic level
Stations, quietest to loudest	under 8 dB	Automatic level
Below 300 Hz	under −25 dB	Cut below 300 Hz
Dropouts	none	Larger Listen buffer; better connection
Squelch tails heard	none	Gate Light, then Strong
Pi load	under 25 %	Cut below 300 Hz off
TX speech level · peak	−24 to −12 · −8 to −1 dBFS	Mic gain
Voice above background	over 20 dB	Noise suppression; Noise gate Light
Limiter working	under 20 %	Lower Mic gain
Gate closed while talking	under 10 %	Noise gate a step lower

Transmissions with gaps	none	Larger TX buffer; Wi-Fi
Peak at the Pi	below −1 dBFS	Lower TX gain

Beyond the rows above, the report has:

- **Per minute**: one bar per minute (up to 15) for speech level, noise floor, Pi load and dropouts, each on a fixed scale so two checks can be compared by eye.
- **The Pi**: its temperature and any throttling, sampled every 10 seconds (kept in memory, not saved).
- **Transmit timing**: the push-to-talk round trip from the page, and the time from key to first audio on the Pi.
- **Node cross-check**: the node's own count of key-ups since it started (`rpt stats`) beside the transmissions your phone made. If the format of `rpt stats` isn't recognised, the raw lines are shown instead.
- **Download report** saves the report as a JSON file on the phone, and **Compare with a saved report** loads one and compares it with the current check. Nothing is kept on the Pi. (In Node Talker Mobile 0.3.1 the download doesn't work; use **Copy report**.)

> **NOTE** The check keeps only numbers, never audio, and only in this browser. It carries on if you switch tabs or reload; time with the page closed or the phone asleep is shown as paused.

8.5 The tone test

The Audio check measures real speech; the tone test measures the path itself. On the Audio check card, **Run tone test** sends your station ID, a series of tones and the ID again through node 1921, listens to what comes back from the node, and shows the response tone by tone: raw from the node, and after the Pi's RX cleanup. Use it to find out which frequencies the radio path passes, what the cleanup does to them, and how long the round trip takes.

SETTING	WHAT IT DOES
Tone length	0.5, 1 (default) or 2 seconds for each tone.
Tones (Hz)	Blank uses 150, 300, 500, 1000, 1500, 2000, 2500, 3000 and 3400 Hz. Type your own list: up to 12 tones, each 100 to 3800 Hz.

With the default settings the test keys the node for about 20 seconds: half a second of quiet (the noise floor), a Morse station ID (*DE* and your callsign), the tones at −10 dBFS with a pause between, then the ID again. Your callsign comes from the Digital card, or failing that from `USRP2M17.ini` or the Dashboard; the test refuses without one.

> **CAREFUL** The tone test transmits. It goes out over the air and to every link on the node, like a Test through the node. It runs from the local network only (Run and Stop), refuses while the channel has audio above the activity level or the node is already transmitting, and is limited to 60 seconds and to the TX time-out less 5 seconds.

The report shows a one-line summary (whether a return was heard and its delay, the usable band, the gains in force and the raw noise floor) and a table with one row per tone:

COLUMN	MEANING
Sent dB	The level sent, after TX gain.
Raw dB · Raw vs sent	What came back from the node before cleanup, and how far that is from what was sent.

After cleanup dB · Cleanup effect	The same tone after the Pi's cleanup (measured 80 ms later when the gate is on, its delay), and the difference from raw.
Freq err Hz	How far the returned tone is from the one sent.
Drops	Breaks in the returned tone.

The *usable band* is where the raw response stays within 6 dB of the 1 kHz tone (or of the strongest tone if 1 kHz isn't in your list). Warnings appear under the table, for example when nothing came back above the noise floor or when the delay can't be trusted. The report stays in memory on the Pi until the next test and is never saved to disk.

8.6 Common mistakes

- **Two automatic gains.** The browser's Automatic gain and the leveler fight; leave the browser's off.
- **Mic gain into the limiter.** More mic gain past the point where the limiter works most of the time adds distortion, not loudness. If the report says the limiter works more than 30 % of the time, turn it down.
- **Fixing the radio in Node Talker.** RX gain and Automatic level only change Node Talker's listeners; a bad radio level still goes to every link.
- **Radio voice on a bright radio.** If the radio adds pre-emphasis, Radio voice makes it shrill.
- **Judging levels through the speaker.** Phone volume and speakers vary; trust the meters.

Reading the spectrum and the meters

The spectrum shows what the node passes, the meters show levels, and the test reports say what to change.

9.1 The spectrum

About 96 bars across, 0 to 4 kHz (marks at 300 Hz, 1, 2 and 3 kHz), redrawn about 25 times a second (Figure 3.1). A bar's height and colour are its loudness on a decibel scale: dark blue is quiet, teal moderate, amber and red loud. The strip below 300 Hz is shaded. Because the scale is in decibels, even a faint sound draws a visible bar: what matters is how the bars *move*. (Before 0.6.12 the spectrum could also be a waterfall; it is bars only now.)

WHAT YOU SEE	WHAT IT IS	WHAT TO DO
Bars rising and falling between 300 Hz and 3.5 kHz with the words, a hump or two around 500–1,500 Hz, the shaded strip quiet	Good speech	Nothing.
One bar in the shaded strip that stands still while a station talks, gone when it stops	A CTCSS (PL) tone	Cut below 300 Hz (Audio levels).
A few bars in the shaded strip (60, 120, 180 Hz or 50, 100, 150 Hz) that stay up between transmissions too	Hum	Cut below 300 Hz; on the node, check the radio interface's grounding.
Every bar across the whole width jumps up at once for a moment after a station unkeys	A squelch tail (hiss)	Squelch-tail gate Light, then Strong.
The bars stay well up across the whole width all the time	Noise: an open squelch, a noisy link	The radio's squelch; the gate.
Bars high up to 3–4 kHz on loud speech, a flat top on the level bar	Clipping (overdriven audio)	Lower the gain where it starts (chapter 8).

9.2 The live meter and the mic test report

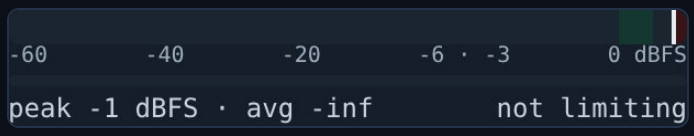


Figure 9.1 · Screenshot. The live meter during a test.

The bar runs from −60 to 0 dBFS. The teal fill is the average, the amber mark the peak, and the thin white mark the highest peak of the last second. The green zone (−6 to −3 dBFS) is where peaks belong; above −1 dBFS is red. The thin bar underneath shows how far the leveler's limiter is turning you down; the text says *limiting 4.2 dB* or *gate closed*.

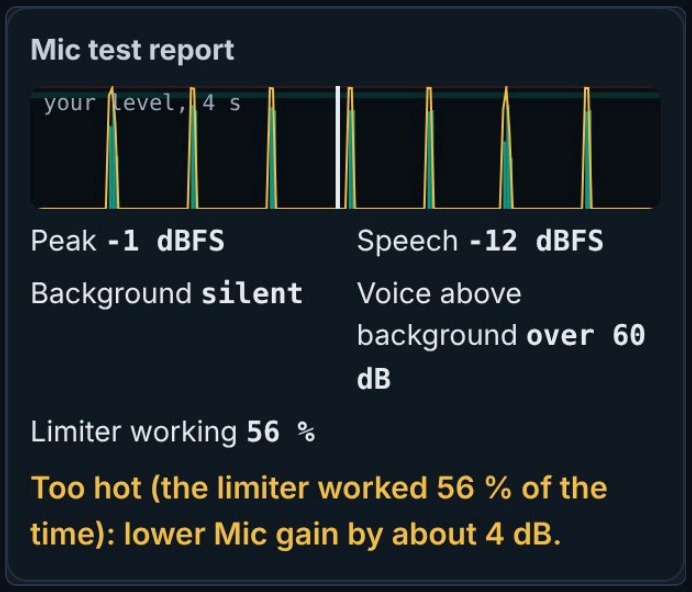


Figure 9.2 · Screenshot (fake test mic). The mic test report: the level over 4 seconds (with a playhead during playback), the numbers, and what to change.

NUMBER	MEANING
Peak	The loudest instant. Should be −8 to −1 dBFS.
Speech	The average while you talk. Should be −24 to −12 dBFS.
Background	The quietest tenth of the recording: the room. *silent* means digital silence.
Voice above background	Speech minus background. Over 20 dB is clean; under 15 dB sounds noisy.
Limiter working	How much of your speech the limiter was turning down. Over 30 %: too much mic gain.
Gate closed while talking	Shown with the noise gate on. Over 10 %: the gate is eating words.

9.3 The node test report

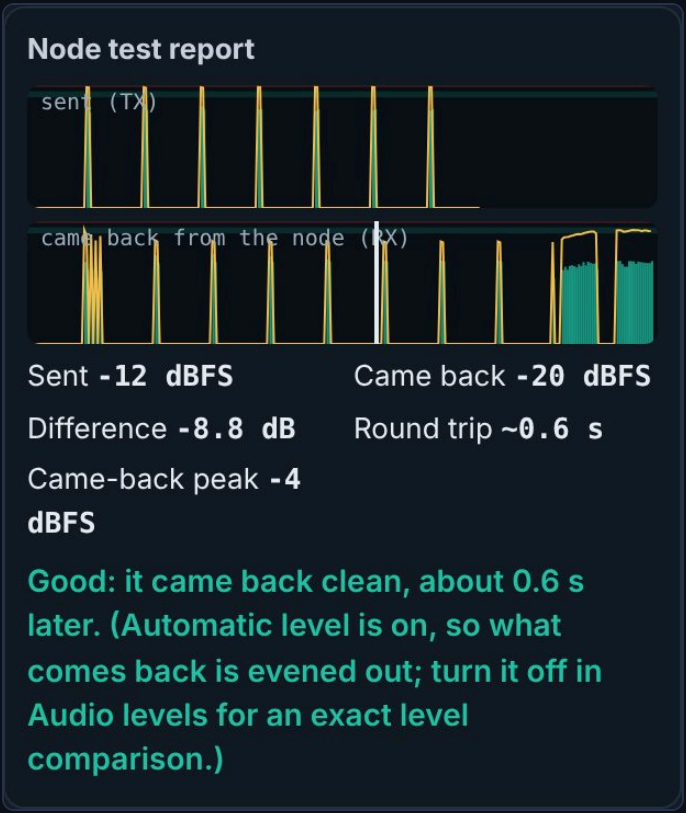


Figure 9.3 · Screenshot (test node). What was sent and what came back, lined up in time.

The test transmits for 4 seconds through the node and records what the node sends back out. The two strips are lined up so the same words sit above each other; **Round trip** is how long that took (about half a second at home). **Difference** compares the speech levels: within about 6 dB is fine.

With Automatic level on, what comes back has been evened out, so the report says so and doesn't judge the difference; turn Automatic level off for a minute to compare exactly.

A *came-back peak* near 0 dBFS means something on the node's side is clipping: TX gain, or the radio and DVSwitch levels in SysMon.

9.4 The level bar while you talk

While you hold push-to-talk, the level bar under the spectrum shows your mic level in red, labelled **TX**; when you let go it goes back to the node's level. Peaks should reach the green zone of the live meter, about −6 to −3 dBFS, on normal speech.

PART III

Everything else

Links and favorites, the Android app, fixing problems, reaching Node Talker from outside, and the reference tables.

10. Connect, Tune and favorites
11. Node Talker Mobile
12. Troubleshooting by symptom
13. From the internet, and security
14. Reference

Connect, Tune and favorites

10.1 Favorites

Setup tab, **Favorites** card: up to 15. **Add row**, pick the mode, type the number and a name, and **Save favorites**. **From the Dashboard...** copies one of the Dashboard's favorites (a DMR one brings its network). They are kept in `/etc/asl_dvs/nodetalker_favorites.conf`, in the Dashboard's `asl_dvs.conf` format, and listed on the Connect tab in that order.

MODE	TYPE
ASL	The node number, up to 7 digits.
EchoLink	The station's own number, up to 6 digits.
DMR	The talkgroup, and its network from the list on the Digital card.
STFU, YSF, P25, NXDN	The talkgroup or reflector, as DVSwitch tunes it.
D-STAR	The reflector and module: `REF001 C` (kept as the tune string `REF001CL`).
M17	The reflector and module: `M17-USA C`, or `M17-USA @ALL` (just `M17-USA` means every module).

Gain per favorite. Each row has two small boxes, **RX dB** and **TX dB** (whole decibels; blank means use the global setting). While that favorite is the active link, its values replace the global **Node to phone (RX)** and **Mic to node (TX)** gains, even with **Adjust audio levels here** off. They go back to the global values when you disconnect the favorite, connect another one, or restart Node Talker. Use them for a net that is always quieter or louder than the rest. They are kept in the favorites file as `|rx=` and `|tx=` after the row.

10.2 The Digital card and DMR networks

The **Digital** card on the Setup tab shows who does the tuning (Solo, or the Dashboard), what was found (`dvswitch.sh`, Analog_Bridge, USRP2M17, `M17Hosts.json`) and the gateway services. It sets the digital and M17 bridge nodes and the M17 callsign; left empty, they are found on their own (the bridge nodes from the USRP ports in `rpt.conf`, the callsign from `USRP2M17.ini`). **DMR networks** holds up to 10 networks, each a name, `address:port`, password and optional dashboard link; **Copy from Node Controller** adds the Dashboard's. The page never shows a password: leave it empty to keep the saved one. **Favorites file**, on the Favorites card, downloads or uploads `nodetalker_favorites.conf`; an upload with a bad line is refused whole.

10.3 Connecting

Connect tab: pick a favorite and press **Connect**; it turns into **Disconnect** while that favorite is on. The page switches to the Listen tab and says *Connecting to ...* or *Tuning ...* under the push-to-talk button; if it fails, it goes back to the Connect tab with the reason. Links are always made from Node Talker's main node, **one at a time**: connecting an ASL or EchoLink favorite drops the node's other public links first, and tuning a digital favorite drops every link but that mode's bridge node. Private nodes (1999 and below: node 1921, the bridge nodes, a private hub) are never dropped, and permanent links are dropped with `ilink 11`. For ASL and EchoLink, Node Talker watches the node's links for up to 15 seconds and says how it went:

RESULT	MEANING
Node 41234 is connected to ASL 2000.	The link is up.
ASL 2000 has not answered after 15 s; node 41234 is still trying.	The far node may be offline or busy.
Node 41234 did not connect to ASL 2000: no link after 15 s.	Check the number; the far node may be refusing connections.

With the Dashboard on the node (its node is Node Talker's main node), every favorite goes through it, so the Dashboard switches to that mode and shows it: ASL, EchoLink and M17 (Dashboard 9.3.89 or later) and the DVSwitch modes (Dashboard 9.3.94-pi02w or later; an older Dashboard gets *update Node Controller*). When the Dashboard has no DMR network for a DMR favorite, the Connect tab asks: **Add it to Node Controller and tune**, **Tune this once** (the Dashboard joins it for this tune only), or **Cancel**.

Solo (the Dashboard not running), Node Talker tunes on its own, the way the Dashboard does. A DVSwitch favorite starts the mode's gateway if needed, runs `dvswitch.sh tune disconnect`, `mode`, for DMR `tune password@address:port` (skipped when that network is already joined), then `tune` the talkgroup, and links the digital bridge node. An M17 favorite writes `USRP2M17.ini` with the address from `M17Hosts.json`, restarts `usrp2m17` and links the M17 bridge node. One digital favorite is on at a time; tuning one drops linked ASL and EchoLink favorites.

10.4 The Tune card

For anything that isn't a favorite, laid out like the Dashboard's search cards (Figure 3.3): **Mode** · **Network** (DMR) or **Module** (D-STAR, M17) · **Search** · the list · **LH** · **Save** · **Connect**, wrapping onto more lines on a phone. **Connect** turns into **Disconnect** while what you picked is linked.

MODE	THE LIST COMES FROM	SEARCH BY
ASL	`/var/lib/asterisk/astdb.txt`, searched on the Pi as you type (2 characters or more, up to 200 matches)	Node number (a number alone matches nodes starting with it), call or place
EchoLink	Asterisk's station list (`echolink dbdump`, kept 5 minutes)	Number or call
DMR	No list: type the talkgroup; pick the network	
STFU	`TGList_BM.txt`	Talkgroup, ID or name
YSF	`YSFHosts.txt`	Name, number or address
P25, NXDN	`P25Hosts.txt`, `NXDNHosts.txt`	Talkgroup or host
D-STAR	The DPlus, DExtra and DCS host lists	Reflector (`REF001 C`) or callsign; module from the Module menu
M17	`M17Hosts.json`	Reflector; the Module menu offers that reflector's modules and All

- Every word typed must match; when one entry is left, it is picked. A talkgroup, room, reflector or node number can also be typed when it isn't in the list, or when the list is missing.
- **Connect** works as for a favorite, through the Dashboard when it runs; a DMR network the Dashboard lacks gets the same add / tune once / cancel question.

- **LH** opens Last Heard: the network's link for DMR, the reflector's own link or address, or the AllStar stats page for an ASL node. Greyed out when there is none.
- **Save** adds the pick to the favorites (up to 15). **Disconnect** (the Connect button while the pick is linked) ends what DVSwitch or M17 has tuned, or, for ASL and EchoLink, drops the link to the node picked or typed. The card shows what is tuned.
- The card starts on the mode last tuned (ASL when there is none) and keeps your mode, network, module, search and pick while the page stays open; its spacing (gap, height, text) is set in the Setup card.

10.5 Links and Disconnect

The **Links** card at the top of the Setup tab lists the node's links, one per line, with the favorite's name. Node 1921 is marked (TX). On the Connect card, **Disconnect** drops that favorite only; other links stay.

Node Talker Mobile (Android app)

An optional Android app that opens only your Node Talker page and runs the audio itself, so receive and transmit keep going with the screen off or while you use other apps. Everything else is the same page you know from the browser.

11.1 Browser or app

	IN A BROWSER	NODE TALKER MOBILE (APP AUDIO)
Screen off, other apps	Audio stops when the page goes to the background	Receive keeps playing; a tap-to-talk transmission carries on
Microphone	Needs the `https://` page	Works on `http://` too
Certificate warning	Each browser warns about the Pi's own certificate	Asked once, by fingerprint; no warnings after that
Phone calls	Depends on the browser	Receive pauses for the call, transmit unkeys
Other apps' sound	Depends on the browser	Plays alongside music or video
Spectrum on the Listen tab	Yes	Empty (the app plays the sound, not the page)
Download report (Audio check)	Saves a JSON file on the phone	Not yet: use Copy report
Microphone processing	The page's leveler, filters and gate (section 6.1)	Android's echo and noise filter, plus the page's Mic gain

11.2 Install and update

1. On the phone, open the repository's Releases page (`github.com/kd8pgk/ASL-DVS/releases`), take the newest `android-v…` release and download `NodeTalker-…-debug.apk`.
2. Open the file. Android asks once to allow installs from the browser; Play Protect may warn about an unknown app: **More details → Install anyway**.
3. Updates install over the old app and keep your nodes and settings. Going to 0.3.1 from an older build needs one uninstall first (the older builds each had their own signing key).

> **NOTE** App audio needs Node Talker 0.6.10 or later for receive and 0.6.11 or later for transmit on the node. With an older Node Talker the app still opens the page and uses page audio.

11.3 Nodes and the certificate

The first screen lists your nodes. **Add node**: a name, the Pi's address (`192.168.1.50` or `node.local`), port `8993`, and HTTPS. A full address such as `https://node.local:8993/` can be pasted into the address field. Tap a node to open it; press and hold to edit, delete or forget its certificate.

- **HTTPS by itself.** If you save a node as plain HTTP and its Node Talker uses HTTPS, the app follows Node Talker's switch to HTTPS and saves the node that way.

- **The certificate.** The first HTTPS connection shows the certificate's SHA-256 fingerprint. Compare it on the Pi with `openssl x509 -noout -fingerprint -sha256 -in /etc/asl_dvs/nodetalker_tls/cert.pem` and tap **Trust**. The app remembers it and stops with a warning if it ever changes (for example after Node Talker makes a new one); trust the new one only if you expected that.
- **Only your node.** The app shows nothing but the node's page. Links to anywhere else (LH, Node Controller, InstMon) open in the phone's browser.
- **Log in** on the page as usual; the app uses the same login for its audio.

11.4 Audio settings

On the node list, **Audio settings**:

SETTING	DEFAULT	WHAT IT DOES, AND WHEN TO CHANGE IT
Audio	App audio	**App audio**: the app plays and records, in the background too. **Page audio**: the page does it, as in a browser, and stops when the screen goes off. Use page audio only if app audio sounds bad.
Receive buffer	300 ms	100 to 1000 ms: how far behind live the app plays. Raise it if audio breaks up (mobile data, weak Wi-Fi); lower it for less delay.
Output	Speaker	**Speaker** uses the media volume; **Earpiece** the call volume, held to the ear like a phone call.
Phone's echo and noise filter	on	Android's own processing on the microphone. Turn off if your transmit audio sounds choppy or muffled.

Also on the node list: **Battery** opens Android's setting that lets the app run in the background (choose *Unrestricted* or *Don't optimize*; the list warns until it is set), and **Keep the screen on while a node is open**.

11.5 How the chain changes in the app

The chain of chapter 4 stays the same on the Pi; only the end on your phone moves from the page into the app:

- **Receive.** When you press Connect feed, the app opens `/api/listen?fmt=s16` itself, in a background service with a notification, and plays it through Android's audio with its own receive buffer. It keeps Wi-Fi awake and reconnects by itself (after 1 s, then up to every 10 s) if the stream drops. The page's Connect feed button, volume slider, level bar and message line follow the app.
- **Transmit.** The app opens the talk channel `/api/talk` itself (with the same login), so the page's push-to-talk button keys through the app. It records the microphone at 8 kHz (or at 16 or 48 kHz and reduces it), applies the page's Mic gain, and sends 20 ms frames, holding up to half a second until the node answers *keyed*, as the page does. The page shows the node's answers (keyed, busy, time-out) and the TX level, unchanged.
- **The guard timer** is kept by the app too, because a page's timers stall with the screen off.

11.6 In the background

WHAT HAPPENS	RECEIVE	TRANSMIT
Screen goes off, or you switch apps	Keeps playing	Tap-to-talk keeps going (until you tap again or the guard timer); hold-to-talk unkeys
Another app plays music or video	Keeps playing, mixed with it	Not affected

Phone call or voice assistant	Pauses, comes back after	Unkeys
The talk channel drops	Not affected	Unkeys; the channel reconnects by itself
App swiped away from recent apps	Keeps playing until **Stop** in the notification	Channel stays ready

The notification shows the node and *Listening*, *Reconnecting…*, *Paused for a call*, *Ready to talk* or *Transmitting*. **Stop** ends listening and the talk channel; tapping it opens the node. Receive is muted while you transmit and comes back 1 s after.

11.7 When something is wrong

SYMPTOM	FIX
The node opens in the phone's browser	Update the app (0.2.1 and later follow the switch to HTTPS), or edit the node and tick HTTPS.
A new build won't install	Uninstall once when coming from a build before 0.3.1.
Audio stops with the screen off	Battery button: set the app to Unrestricted. Check that Audio is App audio.
Audio breaks up	Raise the Receive buffer.
Push-to-talk asks for the microphone	Allow it, then press push-to-talk again.
Log in on the page first	Log in on the page; or turn on Listen without logging in (Setup card) for listen-only use.
Transmit sounds choppy or muffled	Turn the phone's echo and noise filter off (Audio settings).
Certificate changed	Compare the new fingerprint on the Pi; trust it only if Node Talker made a new certificate.

> **NOTE** Still to come: push-to-talk on the volume keys, the app's own settings in the Phone card (section 7.9), saving Audio check reports, and Bluetooth headset microphones and buttons. Test builds use a debug key; when signed release builds start, uninstall the test build once more.

Troubleshooting by symptom

Each from the quickest check to the bigger fix. The Log card and SysMon's Node Talker card (its ASL-DVS tab) show what Node Talker and the node are doing.

No audio at all

1. Status line: *no feed yet* means Asterisk isn't sending. RX card, Node receive: is the line on your node? Send its audio here, then Restart Asterisk.
2. Feed green but silent: is the RX light on? Is Listen showing *Listening.*? Check the volume and the phone's own volume.
3. *Tap anywhere to start listening*: the browser is holding sound until you tap.
4. *All 3 listener slots are in use*: raise Max listeners or wait.

Audio breaks up or stutters

1. Raise the Listen buffer a step (0.5 s on mobile data).
2. Turn 16-bit audio off to halve the data.
3. Move closer to the Wi-Fi access point. The Audio check counts dropouts.

Some stations are much quieter than others

Turn Automatic level on (Audio levels). If one station is quiet everywhere, it is that station.

Low hum or a tone under every transmission

Turn on Cut below 300 Hz. If the spectrum shows hum between transmissions too, it is on the node: check the radio interface's grounding.

A burst of hiss after every transmission

Set the Squelch-tail gate to Light, then Strong. The real fix is the radio's squelch or COS on the node sending it.

My transmissions are too quiet, too loud or distorted

Run the mic test and follow it (section 8.2). Distorted with a lot of limiting: lower Mic gain. Fine in the test but bad on the air: the node test, then TX gain or SysMon's radio levels.

Gaps or choppy audio in my transmissions

Raise the TX buffer to 200 ms; use Wi-Fi rather than mobile data. The Log shows gaps per transmission.

Push-to-talk is refused

MESSAGE	FIX
The node is busy	Wait for quiet, or turn off the busy lockout if a constant noise keeps the node busy.
Someone else is transmitting	Wait; the button shows Busy with who and how long.
Node 1921 is not set up / not linked yet	TX card, Transmit node: Add node 1921, Restart Asterisk, wait 20 s.

Tone test: the channel is busy / a callsign is needed / the test would last too long	The test waits for 3 s without audio above the activity level on the node, so wait for quiet (a constant noise above that level keeps it refusing). It identifies itself on the air, so set your callsign on the Digital card. It is also limited to 60 s and to the TX time-out less 5 s: use fewer tones or shorter ones.
Push-to-talk needs the https:// page	Turn HTTPS on and open the `https://` address, or use Node Talker Mobile with app audio (chapter 11).
Microphone: Permission denied	Allow the microphone for the page in the browser's site settings.

Transmission stops by itself

Time-out: raise TX time-out. *Audio stopped reaching the node*: the phone's connection dropped; use Wi-Fi or raise the TX buffer.

Phone-specific

- **iPhone:** sound needs one tap after the page opens; the lock screen keeps playing while the page is the active audio. Push-to-talk stops when the screen locks.
- **Android:** in a browser, audio stops in the background. Use Node Talker Mobile (chapter 11), and set it to run in the background with its Battery button.
- **Bluetooth headsets** switch to a low-quality call mode when the microphone opens; expect duller listen audio while you test or talk.

From the internet, and security

Node Talker is meant for your home network. Internet access is off unless you turn it on.

- **`allow_wan`** in `/etc/asl_dvs/nodetalker.conf` (no checkbox; then `sudo systemctl restart nodetalker`). Off, clients from the internet get *not open to the internet* for every page. Home addresses, link-local and Tailscale (100.64.0.0/10) always count as home.
- A Cloudflare Tunnel or Tailscale Funnel counts as the internet, whatever address it arrives from.
- From the internet the password is only accepted on the `https://` page.
- On top of the per-address lockout, 20 failed internet logins in 10 minutes make every internet login wait until that window ends. Home logins are never held.

> **CAREFUL** The login is the Pi's root password. Before opening Node Talker to the internet, make sure it is long and used nowhere else, and keep the Dashboard and SysMon on the home network.

Reference

14.1 Files

PATH	WHAT
`/usr/local/lib/asl_dvs/nodetalker/`	The program (linked from `/usr/local/bin/nodetalker.py`; `/usr/local/bin/nodelisten.py` links to that)
`/etc/asl_dvs/nodetalker.conf`	Node settings (14.4), under `[nodetalker]` and a copy under `[nodelisten]`
`/etc/asl_dvs/nodetalker_favorites.conf`	Favorites, DMR networks, bridge nodes and callsign (mode 0600)
`/etc/asl_dvs/nodetalker_favorites.json`	A copy of the ASL, EchoLink and M17 favorites for Node Listen
`/etc/asl_dvs/nodetalker_tuned.json`	What Node Talker tuned on its own
`/etc/asl_dvs/nodetalker_tls/`	The HTTPS certificate and key, made on first start
`/etc/asl_dvs/nodetalker_backups/`	`rpt.conf` before each edit, newest 10
`/etc/asl_dvs/nodelisten*`	Links to the files above, for older tools
`/etc/asl_dvs/local_api.key`	Key for calling the Dashboard

14.2 Ports and service

PORT	WHAT
`8993/tcp`	The page (HTTPS; a plain-HTTP request is redirected)
`6200/udp`	Node audio in, 127.0.0.1 only
`32018 / 34018 udp`	USRP to and from node 1921, 127.0.0.1 only

Service `nodetalker` (also answers to `nodelisten`), through the shared launcher, with `Restart=always` and `WatchdogSec=60`: it stops pinging the watchdog when the page or the audio pump stops answering, and systemd restarts it. `--uninstall` removes the `outstreamcmd` line and node 1921 and keeps the settings; an install within 30 minutes (an update) puts them back.

14.3 Lines Node Talker writes in rpt.conf

They keep the Node Listen names: older tools find the lines by this text, and `nodelisten.py` links to Node Talker.

```
; nodelisten: sends this node's audio to the Node Listen page (remove it from that page)
outstreamcmd = /usr/bin/python3,/usr/local/bin/nodelisten.py,--feed,6200

; nodelisten: transmit node for the Node Listen page (remove it from that page)
[1921](<your node's template>)
rxchannel = USRP/127.0.0.1:34018:32018
duplex = 0
hangtime = 0
althangtime = 0
holdofftelem = 1
telemdefault = 0
nounkeyct = 1
idrecording = |t(0,0,10,0)

[nodes]
1921 = radio@127.0.0.1/1921,NONE
```

14.4 Settings keys (nodetalker.conf)

KEY	DEFAULT	RANGE	PAGE
port	8993	1024–65535	Setup card: Web port
udp_port	6200	1024–65535	Setup card: UDP port
node	(empty)	node number	RX card: Send its audio here
max_listeners	3	1–10	Setup card
public_listen	false	bool	Setup card: Listen without logging in
https	true	bool	Setup card: HTTPS
tot_sec	120	30–600	Setup card: TX time-out
busy_lockout	true	bool	Setup card
tune_enabled	false	bool	TX card: Adjust audio levels here
tx_gain_db / rx_gain_db	0	−12–12	Audio levels (RX and TX cards)
jitter_ms	250	100–1500	RX card: Listen buffer
tx_buffer_ms	120	40–400	TX card: TX buffer
rx_hpf / rx_agc	true	bool	RX card: RX cleanup
rx_gate	1	0 off, 1 light, 2 strong	RX card: Squelch-tail gate
ptt_mode	hold	hold, tap	TX card: the copy of Push-to-talk on this phone
ptt_guard_sec	60	30, 60, 90, 120, 180	TX card: Guard timer (copy)
ptt_rx_gap	false	bool	TX card: Quick Key Guard (copy)
allow_wan	false	bool	none (chapter 13)

14.5 API

REQUEST	WHAT
`GET /api/health`	Version; no login. Used by the watchdog and SysMon.
`GET /api/whoami`	Logged in?, public listening, Dashboard and SysMon ports; no login.
`GET /api/status`	Feed, levels, listeners, links, transmit state, Pi load (login, or public listening); the Pi's temperature and throttling only with a login
`GET /api/listen[?fmt=s16]`	The audio stream (login, or public listening)
`GET /api/talk`	WebSocket for push-to-talk (login, same origin)
`GET /api/node` · `/api/tx` · `/api/favorites` · `/api/dash-favorites` · `/api/digital` · `/api/favorites-file` · `/api/config` · `/api/log`	Setup data (login)
`GET /api/node/counters`	The node's key-up count and transmit time from `rpt stats` (login); used by the Audio check's node cross-check
`GET /api/tonetest/status`	Progress and the last report of the tone test (login)
`GET /api/tune-list?mode=` · `/api/tune-search?mode=&q=`	The Tune card's lists and the ASL and EchoLink search (login)
`POST /api/login` · `/api/logout`	Session
`POST /api/config` · `/api/node/enable` · `/api/node/disable` · `/api/tx/enable` · `/api/tx/disable` · `/api/favorites` · `/api/connect` · `/api/disconnect` · `/api/tune` · `/api/tune/save` · `/api/tune/disconnect` · `/api/digital` · `/api/dmr-networks` · `/api/dmr-networks/import` · `/api/tonetest/start` · `/api/tonetest/stop` · `/api/favorites-file` · `/api/asterisk/restart` · `/api/system/reboot` · `/api/system/shutdown`	Changes (login; header `X-Requested-With: XMLHttpRequest`)

14.6 Version history (highlights)

VERSION	WHAT ARRIVED
0.1.0–0.2.0	Listening in the browser through one `outstreamcmd` line
0.3.0	Phone page with tabs, HTTPS, push-to-talk on node 1921, favorites
0.4.0–0.4.5	Tune card (gains, buffers), filtered resampling, Leveler, noise gate, Radio voice, soft limits
0.4.7	Internet access rules (`allow_wan`)

0.4.10	M17 favorites through the Dashboard
0.4.11	Connect reports whether the link came up
0.4.12–0.4.13	Listening starts on open; Setup cards open and close; always the main node
0.4.14	One-screen Listen tab with the waterfall; Status tab removed
0.4.15	RX cleanup: Cut below 300 Hz, Automatic level, Squelch-tail gate
0.4.16	Live meter and reports for the mic and node tests; TX on the level bar
0.4.17	Audio check; the squelch-tail gate tells hiss from voice more reliably
0.4.18–0.4.20	Connect turns into Disconnect; headset button and audio output; login and install fixes
0.5.0	15 favorites with DMR, STFU, YSF, P25, NXDN and D-STAR; DMR networks; tuning on its own; the Digital card
0.6.0	Renamed Node Talker; DVSwitch favorites through the Dashboard; add a DMR network from the Connect tab
0.6.1	One link at a time, as the Dashboard does; private nodes left alone; the link guard
0.6.2–0.6.4	Tap to talk with a guard timer; connect reports on the Listen tab; wait 4 s after receive
0.6.5–0.6.7	System card (Reboot, Shutdown); favorites shown by name
0.6.8–0.6.9	Tune card (every mode, with ASL and EchoLink search); Links card moved to the Setup tab
0.6.10–0.6.11	Receive and transmit through Node Talker Mobile (app audio)
0.6.12	Spectrum is bars only
0.6.13–0.8.x	No per-version history was kept for these. Changes between 0.6.12 and 0.9.0 that the notes of 0.9.x do not cover include gain per favorite, Quick Key Guard (was Wait 4 s after receive), the Connect feed button and the regrouping of the Setup tab into its present cards
0.9.0–0.9.3	Tone test (Pi side, then in the page)
0.9.4–0.9.7	Audio check: dropout classes, per-minute strip, Pi temperature, PTT round trip, node cross-check, report download and compare
0.9.8–0.9.12	Tone test busy check fixed; Favorites and Digital become cards of their own; Copy all on every card; `rpt stats` reads the count since start-up
0.9.13–0.9.17	The link line shows the live link (ASL node, DVSwitch or M17 bridge)
0.9.20	RX cleanup: gate tells hiss from tones, keeps 40 ms before opening and fades in; TX fades at a gap; Phone mic filter
0.9.21	Settings that can't be used are grayed out; the Phone card shows the app
0.9.22	The first build in the repository: `ptt_rx_gap` off by default; Pi health only with a login; the tone test is local-only for Stop too and limited by the TX time-out; node 1921 gets a silent ID

NODE TALKER MOBILE	WHAT ARRIVED
0.1.0	Node list, the page alone, certificate trusted by fingerprint
0.2.0–0.2.1	Receive in the background; follows the switch to HTTPS
0.3.0–0.3.1	Transmit in the app; background play everywhere; Battery button; one signing key; renamed Node Talker Mobile

The full lists are in `changelogs/nodetalker.md` and `changelogs/nodetalker_android.md`.